

# Topic 16: Multiple Random Variables

02-680: Essentials of Mathematics and Statistics

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Sometimes we need to deal with the interactions of more than one random variable.

Consider tossing a (fair) coin 3 times, and define two random variables:

- $X$  — the number of heads in the first toss
- $Y$  — the number of heads in all 3 tosses

We want to know the joint probability (that is, the probability  $p(X = x, Y = y)$ ):

|   |   | Y   |     |     |     |
|---|---|-----|-----|-----|-----|
|   |   | 0   | 1   | 2   | 3   |
| X | 0 | 1/8 | 1/4 | 1/8 | 0   |
|   | 1 | 0   | 1/8 | 1/4 | 1/8 |

## 1 Marginal Probabilities

What if we're given the joint probability of some events and want to find out the underlying probability of the events.

**Example.** Assume we have two *unfair* coins, which we will model as random variables  $X$  and  $Y$ ,

and we're given the following joint probabilities:

|   |       | Y     |       |      |
|---|-------|-------|-------|------|
|   |       | Heads | Tails |      |
| X | Heads | 1/10  | 2/10  | 3/10 |
|   | Tails | 3/10  | 4/10  | 7/10 |
|   |       | 4/10  | 6/10  |      |

If we want to determine, say  $P(X = \text{Tail})$ , it turns out we can sum over all the possibilities of  $Y$ :

$$P(X = \text{Tail}) = \sum_{y \in \{\text{Head}, \text{Tail}\}} p(X = \text{Tail}, Y = y) = \frac{3}{10} + \frac{4}{10} = \frac{7}{10}$$

Doing all the math, both coins are biased toward **Tails**, the first coin ( $X$ ) more-so.

## 1.1 Continuous Random Variables

For continuous random variables, this again turns from sums into integrals. Assume you're given a joint distribution function  $f(x, y)$ . We can, as before, find the joint probability for a range as a (double) integral:

$$p(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x, y) dy dx$$

And to get a marginal distribution function

$$f(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

**Example.** Lets say we're drawing points uniformly over the unit square, in this case

$$f(x, y) = \begin{cases} 1 & 0 \leq x, y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

So if we want to find

$$\begin{aligned} p\left(X \leq \frac{1}{2}, Y \leq \frac{1}{2}\right) &= \int_0^{\frac{1}{2}} \int_0^{\frac{1}{2}} 1 dy dx = \left((xy)|_0^{\frac{1}{2}}\right)\bigg|_0^{\frac{1}{2}} = \left(x \cdot \frac{1}{2} - x \cdot 0\right)\bigg|_0^{\frac{1}{2}} \\ &= \left(\frac{1}{2} \cdot \frac{1}{2} - \frac{1}{2} \cdot 0\right) - \left(0 \cdot \frac{1}{2} - 0 \cdot 0\right) = \frac{1}{4} \end{aligned}$$

## 2 Independence of Random Variables

Similar to independence for event probabilities, two random variables  $X$  and  $Y$  are independent iff

$$\forall x, y : p(X = x, Y = y) = p(X = x) \cdot p(Y = y)$$

Work left to the reader: one of the situations below describes independent random variables the other does not.

|   |       | Y     |       |     |
|---|-------|-------|-------|-----|
|   |       | Heads | Tails |     |
| X | Heads | 1/4   | 1/4   | 1/2 |
|   | Tails | 1/4   | 1/4   | 1/2 |
|   |       | 1/2   | 1/2   |     |

|   |       | Y     |       |     |
|---|-------|-------|-------|-----|
|   |       | Heads | Tails |     |
| X | Heads | 1/2   | 0     | 1/2 |
|   | Tails | 0     | 1/2   | 1/2 |
|   |       | 1/2   | 1/2   |     |

### 3 Conditional Distributions

Similar to independence for event probabilities, for two random variables  $X$  and  $Y$  :

$$p(X | Y) = \frac{p(X, Y)}{p(Y)}$$

| BRCA1 mutation | P53 mutation | develops cancer |                  |
|----------------|--------------|-----------------|------------------|
| $G_1$          | $G_2$        | $C$             | $p(G_1, G_2, C)$ |
| N              | N            | N               | 0.24             |
| N              | N            | Y               | 0.01             |
| N              | Y            | N               | 0.20             |
| N              | Y            | Y               | 0.05             |
| Y              | N            | N               | 0.15             |
| Y              | N            | Y               | 0.10             |
| Y              | Y            | N               | 0.05             |
| Y              | Y            | Y               | 0.20             |

Compute  $p(C | G_1, G_2)$ .

$$p(C | G_1, G_2) = \frac{p(G_1, G_2, C)}{p(G_1, G_2)} = \frac{p(G_1, G_2, C)}{p(G_1, G_2, C = N) + p(G_1, G_2, C = Y)}$$

Note that

$$p(G_1 = a, G_2 = b) = \frac{\sum_{c \in \{Y, N\}} p(G_1 = a, G_2 = b, C = c)}{\sum_{g_1 \in \{Y, N\}} \sum_{g_2 \in \{Y, N\}} \sum_{c \in \{Y, N\}} p(G_1 = g_1, G_2 = g_2, C = c)}$$

but because the denominator is 1 it is left out of the equation above for ease exposition.

| $G_1$ | $G_2$ | $p(C = Y   G_1, G_2)$           |
|-------|-------|---------------------------------|
| N     | N     | $\frac{0.01}{0.24+0.01} = 0.04$ |
| N     | Y     | $\frac{0.05}{0.20+0.05} = 0.20$ |
| Y     | N     | $\frac{0.10}{0.15+0.10} = 0.40$ |
| Y     | Y     | $\frac{0.20}{0.05+0.20} = 0.80$ |

## Useful References

Wasserman. “All of Statistics: A Concise Course in Statistical Inference” §§2.5-2.7